

ERP System Implementation Project (Odoo)

Watan Designs Limited, Kingdom of Saudi Arabia

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For Digital Marketing and eCommerce Services

Odoo Quality Module (Chapter 07)

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7.0 Production Quality Management Control

7.1 Introduction And Objectives

The quality management system is a core component of modern manufacturing processes, aiming to monitor the quality of processes and products at every stage of the production cycle. The system allows for the creation and implementation of detailed inspection points within production lines, recording inspection results, tracking non-conformities, and applying corrective and preventive actions to ensure the product meets the required standards.

System Objectives

- Controlling the quality control cycle at all stages of production.
- Recording the results of the inspection and analyzing the causes of operational defects.
- Ensuring traceability of materials and production batches using barcodes and serial Numbers.
- Automating internal accreditation and audit processes.
- Creating a standard database for inspection and quality reports.
- Improving communication between the quality, design, production and warehouses.

7.2 Basic Terms And Concepts

Term	Translation	Explanation
Quality Control Point (Inspection Point)	نقطة فحص الجودة أو نقطة التفتيش	A stage within the production line where a specific test or inspection is carried out to ensure that the product conforms to the original design and required standards.
Quality Check (Inspection Instance)	فحص الجودة أو عملية التفتيش	The practical procedure in which the actual inspection is carried out, and the result (conforming/non-conforming) is recorded.
Non-conformance (Defect / Non-Conformity)	عدم المطابقة أو العيب المرصود	A status that is recorded when a defect or non-compliance with specifications is discovered, and is used to track and analyze recurring problems.
Corrective Action (Fix Action)	إجراء تصحيحي أو إجراء معالجة	The action taken to address the cause of the defect after a non-conformity is detected (such as rework or process modification).
Preventive Action (Preventive Measure)	إجراء وقائي أو تدبير احترازي	An action taken to avoid repeating the same mistake in the future, such as improving operating procedures or training employees.
Tolerance Range (Acceptable Limits)	حدود السماح نطاق التفاوت	The acceptable margin of error is the deviation in measurements or technical specifications.

Root Cause Analysis (Cause Evaluation)	تحليل السبب الجذري أو رصد مصدر العيب	An analytical process used to identify the underlying cause of a recurring quality problem and suggest appropriate actions.
Quality Alert (Internal Notification)	تنبيه جودة أو إنذار مراقبة	A notification is automatically generated when an inspection fails or a defect is detected, to notify the factory manager.
Quality Tag (Label / Category)	تصنيف الجودة أو بطاقة التصنيف	A mark is added after quality inspection and defect detection to classify and clarify the fate of the products.
Inspection Frequency (Control Interval)	تكرار الفحص أو معدل الفحص	The number of times a particular inspection is carried out (per invoice, per room, or once a day).
Quality Measure (Metric / Indicator)	مقياس الجودة أو معيار الاختبار	The unit of measurement used in the inspection (e.g., length, stiffness, stitch integrity).
Rework Order (Reprocessing Order)	أمر إعادة العمل أو أمر التصحيح	An order created to remanufacture or modify products that do not conform to specs.
Inspection Report (Quality Report)	تقرير الفحص أو نتيجة التقييم	A document generated by Odoo showing the detailed results of the examination, and used as a documentation reference.

7.3 Quality Cycle Workflow

No.	Stage	Explanation
1	Quality Plan (تحديد خطة الجودة)	The quality cycle begins by defining an inspection plan for each order, process, or department, which includes inspection points, types of tests, and acceptable standards.
2	Quality Control Points (إنشاء نقاط الفحص)	The system establishes checkpoints at specific times (pre-production, in-process, and post-production), and associates their execution with the relevant product.
3	Quality Checks (تنفيذ الفحص وتسجيل النتائج)	The inspector conducts the inspection at the specified point and records the result within the system (accepted - rejected - needs review).
4	Non-Conformance (إدارة حالات عدم المطابقة)	A non-conformity report is created upon the discovery of a defect. This report includes specific details about the issue and identifies the party responsible.
5	CAPA (تنفيذ الإجراءات التصحيحية والإجراءات الوقائية)	The root cause of the defect is analyzed, an immediate corrective action or long-term preventive action is recorded, and implementation is monitored until closure.
6	Approval Workflow (اعتماد نتائج الفحص)	After reviewing the results, the supervisor approves the in-system check to ensure that the standards are met.
7	Analysis & Reporting (تحليل الأداء وإصدار التقارير)	Periodic reports are generated to track acceptance and rejection rates, most frequent defects, and to evaluate production performance monthly and annually.

7.4 System Operating Procedures (Step-by-Step)

A. Preparing a Quality Plan (Quality Plan Setup)

1. From the Quality module, select "Quality Control Points" and then click "Create" to create a new plan, such as "Final paint inspection".
2. Determine the appropriate type of inspection (before Production – in-process – post production) to distribute the inspection across the production stages.
3. Enter the target product to ensure the plan is automatically applied when any related manufacturing order is run.
4. In the "Control Type" tab, specify the type of check: Standard (Pass/Fail), Quantitative (Measure), or Visual Reference (Reference).
5. Add the acceptable limits (Tolerance Values) accurately (e.g., length ± 0.5 cm).
6. It is preferable to create "standard quality plan templates" for each department to facilitate reuse and ensure consistency between projects and factory branches.

B. Creating Quality Control Points

1. From the same section, press "New Control Point" to specify the inspection location (stage, machine, or product).
2. Specify the type of operation involved: receiving materials, production, or final shipment.
3. Link the checkpoint to the Work Center.
4. Add detailed instructions in the Instructions box to clarify the inspection steps.
5. Enter the required frequency as "every operation" or "random".
6. Make sure to activate the "Automatically Triggered" option to activate the scan without manual intervention.

7. It is recommended to activate a mandatory check-up in critical stages such as painting, because ignoring a single inspection step could lead to rework and wasted time.

C. Performing Quality Checks and Recording Results

1. When production starts, inspection points automatically appear within the manufacturing screen.
2. The result is entered according to the type of inspection (digital measurement or accept/reject), with the possibility of using a barcode reader to identify the product.
3. When registering "Fail", the "Create Quality Alert" option appears automatically to open an immediate notification of the defect.
4. The production supervisor must attach preview images that document the actual state of the defect, using "Odoo Mobile" application to reduce the chances of error and document the situation in real time.

D. Non-conformance Handling

1. When a non-conforming inspection is recorded, an automatic report is generated in the Quality Alerts section.
2. Enter a detailed description of the problem, and specify the affected product and quantity.
3. Choose the defect category (manufacturing, materials, finishing).
4. Add the person responsible for reviewing the status to the "Assigned To" box, to ensure that implementation is tracked.
5. The system will send an automatic notification to the factory manager as soon as the report is created.
6. After reviewing the case, it is determined whether the product needs to be reworked or scrapped.

7. Make sure to use consistent defect categories such as "inaccurate dimensions" or "incomplete paint" or "color deviation", as this makes it easier to generate trend analysis reports.

E. Implementing Corrective and Preventive Actions

1. From within the defect report, click "Create Corrective Action" to document the required action. Then, in the Root Cause section, document the root cause.
2. Add the corrective action as "replacing the supplier's material" or "adjusting the machine settings".
3. Add a long-term preventive measure such as "training the carpentry team" or "alerting a specific worker".
4. Every CAPA report must be closed within a period not exceeding 72 hours from the date of discovery of the defect, to avoid the accumulation of operational problems.

F. Quality integration with other modules

1. With the Manufacturing Module, an automatic check is created after each manufacturing order, and the order is prevented from being closed before quality approval.
2. With the Inventory Module, when the check fails, the rejected quantity is automatically moved to the "Rejected Products" location.
3. With the Purchasing Module, upon receipt of materials, an automatic check is created to determine whether the order is accepted or rejected.
4. These integrations are what transform the quality system into a part of the entire work cycle and not a separate unit, reducing response time and preventing error repetitions.

7.5 Frequently Asked Questions (FAQ)

Q: Can the system automatically detect defective products during production?

Yes, through inspection points linked to the production lines, the inspector is alerted as soon as a measurement or value is entered outside the specified tolerance range. This allows for immediate correction of defects without waiting for the process to complete or requiring manual intervention.

Q: Can the manager track the performance of the inspectors and the quality of the inspections?

Yes, through the Quality Dashboard, which displays the number of inspections performed and the acceptance and rejection rates for each inspector. This data provides accurate performance indicators that support employee evaluation and improve the efficiency of quality teams.

Q: How can raw materials rejected from quality inspection be dealt with?

Rejected items are automatically moved to a "temporary isolation" location within the inventory until they are reviewed by the purchasing department or supplier. Inspection results and recommendations are recorded in the system to ensure transparent communication between departments.

Q: Can field inspections be documented with photos or files?

Yes, the inspector can attach photos or measurement reports directly to the inspection log from their phone or tablet. This helps document the actual condition of the product, and facilitates subsequent review of any disputes or analyses.

Q: What is the method for tracking the causes of recurring defects?

Each non-conformity report is categorized by defect type and source, and the system generates a periodic report analyzing trends. This helps identify root causes and implement effective preventative measures to reduce recurrence.

Q: Can the system prevent unapproved products from leaving the warehouse?

Yes, when quality is linked to inventory, the transfer of unapproved products is prohibited until they are approved by the quality manager. This integration helps ensure that all outgoing shipments conform to factory standards.

Q: Can the Quality Module generate reports suitable for submission to management or customers?

Yes, inspection reports can be automatically extracted from the system in PDF or Excel formats.

The reports include acceptance rates, details of examinations, corrective actions, and recommendations for future improvement.

7.6 The Quality Supervisor's Daily Checklist

1. Review the tests performed the previous day.
2. Make sure all scheduled checkpoints are recorded.
3. Follow up on new non-conformity reports.
4. Adoption of the final inspection results for finished products.
5. Updating daily quality feedback in the system.
6. Review open alerts and prioritize handling.
7. Prepare a brief daily report on the overall quality status.